```
# PROGRAM 1: SET OPERATIONS

A <- c(1, 2, 3, 4, 5)
B <- c(4, 5, 6, 7, 8)

cat("Union:", union(A, B), "\n")
cat("Intersection:", intersect(A, B), "\n")
cat("Difference A-B:", setdiff(A, B), "\n")
cat("Is A a subset of B?", all(A %in% B), "\n")


# PROGRAM 2: INVERSE FUNCTION

inverse_function <- function(x) {
  return(1 / x)
}
x <- 5
cat("Inverse of", x, "is", inverse_function(x),
"\n")
```

```
# PROGRAM 3: ONE-TO-ONE FUNCTION

one_to_one_function <- function(x) {
  return (2 * x + 3)
}
x_values <- c(1, 2, 3, 4)

cat("Input values:", x_values, "\n")
cat("Function values:",
one_to_one_function(x_values), "\n")



# PROGRAM 4: CARTESIAN PRODUCT OF
TWO SETS

A <- c(1, 2)
B <- c("a", "b")

Cartesian product<- expand.grid(A,B)
cat("Cartesian product of A and B:\n")
print(expand.grid(A = A, B = B))
```

```
# PROGRAM 5: CHECK REFLEXIVE
RELATION

relation <- matrix(c(1, 2, 3,2, 1, 3,3, 2, 1),
nrow = 3, byrow = TRUE)
reflexive <- all(diag(relation) == c(1, 1, 1))
cat("Is the relation reflexive?", reflexive,
"\n")


# PROGRAM 6: CHECK TRANSITIVE
RELATION

relation <- matrix(c(1, 1, 0,1, 1, 0,0, 0, 1),
nrow = 3, byrow = TRUE)
Transitive<-
all((relation%*%relation)<=relation)

cat("Is the relation transitive?", transitive,
"\n")
```

```
# PROGRAM 7: CHECK SYMMETRIC RELATION

relation <- matrix(c(1, 1, 0, 1, 1, 0,0, 0, 1),
                   nrow = 3, byrow = TRUE)

symmetric <- all(relation == t(relation))
cat("Is the relation symmetric?",
symmetric, "\n")


# PROGRAM 8: LOGIC GATES (NOT, AND, OR, XOR)

x <- TRUE
y <- FALSE

cat("NOT(x):", !x, "\n")
cat("AND(x,y):", x & y, "\n")
cat("OR(x,y):", x | y, "\n")
cat("XOR(x,y):", xor(x, y), "\n")
```

```
# PROGRAM 9: MEAN, MEDIAN AND MODE

data <- c(1, 2, 2, 3, 4, 5.5, 5, 6, 7, 8, 9, 19)

unique_values <- unique(data)
frequencies <- tabulate(match(data,
unique_values))
mode_value <-
unique_values[which.max(frequencies)]

cat("Data:", data, "\n")
cat("Mean:", mean(data), "\n")
cat("Median:", median(data), "\n")
cat("Mode:", mode_value, "\n")


# PROGRAM 10: MATRIX OPERATIONS

A <- matrix(c(1, 2, 3, 4), nrow = 2, byrow =
TRUE)
B <- matrix(c(9, 8, 7, 6), nrow = 2, byrow =
```

```
TRUE)

print(A + B)

print(A - B)

print(A %*% B)

print(t(A))

Print( det(A))


# PROGRAM 11: CUMULATIVE SUM,
PRODUCT, MINIMA AND MAXIMA

data <- c(1, 2, 3, 4, 5)

cat("Cumulative Sum:", cumsum(data), "\n")
cat("Cumulative Product:", cumprod(data),
"\n")
cat("Minimum:", min(data), "\n")
cat("Maximum:", max(data), "\n")
```

```
# PROGRAM 12: ARITHMETIC MEAN FOR
UNGROUPED AND GROUPED DATA

# Ungrouped Data

ungrouped_data <- c(2, 4, 6, 8, 10)

mean_ungrouped <-
mean(ungrouped_data)

cat("Arithmetic Mean for Ungrouped Data:",
    mean_ungrouped, "\n")

# Grouped Data

values <- c(2, 4, 6, 8, 10)
frequencies <- c(1, 2, 3, 2, 1)

mean_grouped <- sum(values *
frequencies) / sum(frequencies)
```

```
cat("Arithmetic Mean for Grouped Data:",
    mean_grouped, "\n")


# PROGRAM 13: COEFFICIENT OF
VARIATION

# Discrete Series

discrete_data <- c(5, 6, 7, 8, 9)

sd_discrete <- sd(discrete_data)
mean_discrete <- mean(discrete_data)

cv_discrete <- (sd_discrete /
mean_discrete) * 100

cat("Discrete Series Coefficient of
Variation(%):",
    cv_discrete, "\n")

# Continuous Series
```

```
continuous_data <- c(10.5, 12.3, 11.4, 10.7,
11.9)

sd_continuous <- sd(continuous_data)
mean_continuous <-
mean(continuous_data)

cv_continuous <- (sd_continuous /
mean_continuous) * 100

cat("Continuous Series Coefficient of
Variation(%):",
    cv_continuous, "\n")
```